

Operating a Volunteer OpenSky Node in Brazil: Field Notes on Coverage, Uptime, and Local Regulations

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Abstract

This field report documents a volunteer OpenSky receiver in Franca, Brazil, using local ADS-B traces through September 2026. Replacing a home-built antenna with an AirNav XBoost raised median range from 189.3 to 222.0 km. In matched windows, pooled mean range increased by 15.6% for 2,234 aircraft observed with both antennas. A persistent blind sector suggests local obstruction; 3 exceptional propagation windows exceeded 600 km. Reception continuity was 99.33%, while uptime based on two identified receiver outages was 99.54%. The operator reports power, software and thermal interruption causes without event-level attribution. Regional electricity indicators and an illustrative import-cost calculation contextualise the installation. The findings describe one receiver, not Brazilian or network-wide coverage; archived aggregates and scripts accompany the poster.

Keywords: ADS-B; crowdsourcing; receiver coverage; Brazil; regulation

1. Motivation and installation

The OpenSky Network relies on volunteers maintaining their own receivers. Its user survey identifies cost as a major barrier in under-represented regions [1]. This case study connects antenna performance, operational continuity and practical onboarding constraints at sensor –1408044782 in Franca.

Franca, in north-eastern São Paulo, southeastern Brazil, is among the state's higher-elevation cities, approximately 1,040 m above sea level, with urban hills separated by stream valleys [2]. The antenna is at approximately -20.51° , -47.40° , 1,030 m above sea level and 4 m above ground. A Raspberry Pi 5 runs ultrafeeder (readsb 3.16.14) with an AirNav ADS-B 1090 MHz FlightStick SDR. A home-built antenna was replaced by an AirNav XBoost on 2026-05-16 at 21:35 UTC.

2. Data and coverage

Era A covers 2026-04-13 to 2026-05-14 (32 days); Era B covers the swap to 2026-09-30 (approximately 137 days, including a partial first day). Analysis uses local readsb traces, not OpenSky database records. Only adsb_icao and adsb_icao_nt positions are retained; multilateration is excluded. Ranges use haversine distance ($R = 6371.0088$ km) from the archived rounded origin -20.51 , -47.40 . Its approximately 1 km rounding error is retained for consistency with the archived results [3].

Table 1 separates the full record from equal-length comparisons. In Era A and the paired Era B window, the common fleet of 2,234 aircraft had pooled mean ranges of 188.5 and 217.9 km (+15.6%). The paired window yielded 57.7% more positions and 9.3% more aircraft. Matching aircraft and window length reduces fleet-composition differences, but does not eliminate changes in routes or traffic.

Table 1. ADS-B metrics. Paired Era B spans 32 days after the swap. Both Era B maxima include the 17 June propagation event.

Metric	Era A	B paired	B full
Days	32	32	137
Positions	5,153,167	8,123,970	36,535,422
Aircraft	2,890	3,159	4,925
Median (km)	189.3	224.0	222.0
P95 (km)	299.7	350.6	350.1
Maximum (km)	510.3	786.3	786.3

Figure 1 shows a blind sector from 290° to 350° with both antennas, consistent with terrain or local obstruction. Coverage peaks at 140° to 200°, towards São Paulo. The 3 declared ducting windows with positions beyond 600 km were 16 to 18 June, 15 August and 25 August. Maximum observed range was 786.3 km; excluding the exceptional positions within these windows gives 599.9 km. These observations do not establish the long-term frequency of ducting.

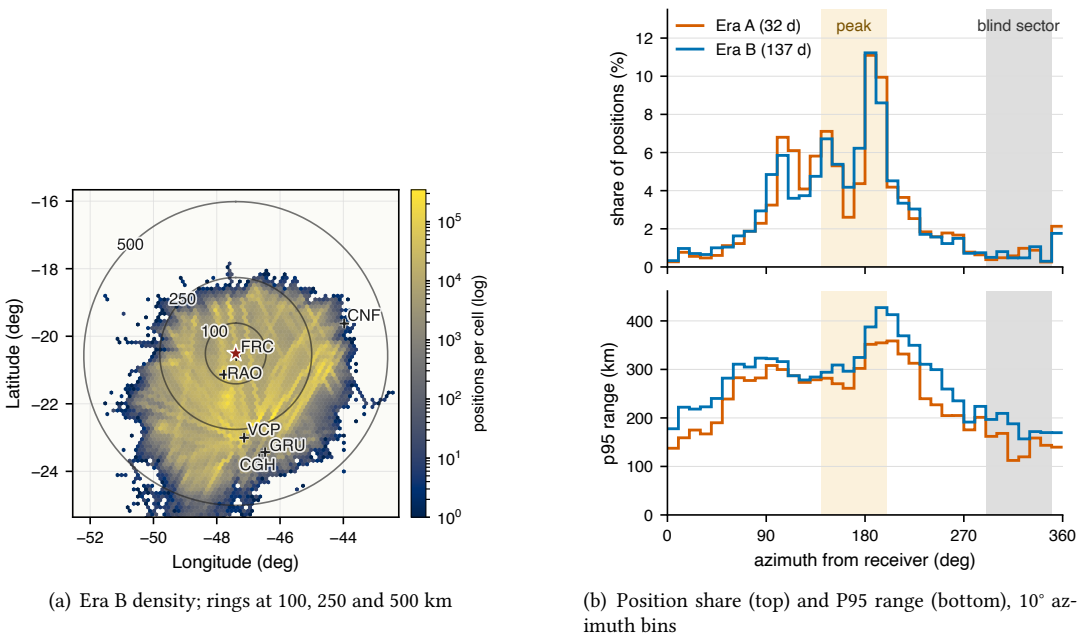


Figure 1. Local ADS-B coverage in Franca. Receiver origin rounded to two decimals. Traffic density is not guaranteed reception coverage.

3. Continuity and operating constraints

Across 3290.4 h, 40 reception gaps above 120 s totalled 21.91 h, giving 99.33% reception continuity. Of these, 19 began between 05:00 and 07:00 UTC; available statistics cannot distinguish receiver failure from an absence of observable traffic. Two identified receiver outages lasted 5 h 12 min on 6 July and 9 h 47 min on 2 August. Counting only these gives 99.54% receiver uptime. A further 64-minute gap on 30 August remains unexplained.

The operator reports occasional power loss, k3s conflicts after updates and USB SDR-stick overheating beyond its operating limit. No event-by-event attribution or cause-specific downtime is available. Reception continuity must therefore be distinguished from receiver uptime and electricity reliability. Regional context also matters: average interruption duration per consumer in 2022 was about 7 h in the Southeast and 24 h in the North [4]; one Southeast installation cannot represent Brazil.

Practical regulatory touchpoints include equipment homologation, home-location privacy and international data transfers; the companion checklist documents these issues [3, 5]. ADS-B-only observations cover the equipped fleet; continental deployment and equipage policies are discussed by DECEA [6]. Import costs add a concrete barrier: the illustrative Remessa Conforme calculation for a US\$150 kit adds US\$60 duty and approximately US\$43 ICMS, grossed up at 17%, for about US\$253 (+69%) [7]. This is a scenario, not a quotation for every purchase.

4. Conclusions and limitations

The antenna change coincided with improved range and message capture, while persistent directional obstruction and mixed interruption causes remain operational constraints. Cost, information and regional electricity reliability merit separate investigation. This single-node record cannot rank national barriers or measure OpenSky coverage. The last 33 days lack a source checksum; export-file hashes do not resolve that provenance limitation. A prior audit with an apparent date-label error is not used.

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Open data statement

Raw receiver traces are openly archived as dataset v1.0.0 under CC BY 4.0, DOI [10.5281/zenodo.23090597](https://doi.org/10.5281/zenodo.23090597) [8]. Aggregate metrics, figures, code and the regulatory checklist are archived as v5.0.0, DOI [10.5281/zenodo.23089559](https://doi.org/10.5281/zenodo.23089559) [3].

Reproducibility statement

Restore the public trace archives into the documented directory structure and run the archived dataset-building and metric scripts to recompute results. Checksums verify the input files; scripts specify time windows and ADS-B filtering. The QR links to dataset v1.0.0.



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